# Reinforcement Learning for Dynamic Operation of a $CO_2$-to-Methanol Plant

Technical Report — extension of the EURECHA Process Design Contest

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**Abstract**

I present a reinforcement-learning framework for the hour-by-hour dynamic operation of a cement-integrated $CO_2$-to-methanol synthesis plant (213 kt/yr, 270 MW PEM electrolyser) under volatile electricity pricing. Three RL algorithms—Soft Actor-Critic (SAC), Proximal Policy Optimisation (PPO), and tabular Q-learning—are benchmarked against rule-based and constant-load baselines. SAC delivers approximately **+490% profit** versus constant full-load operation and generalises zero-shot across electricity markets (GB $\leftrightarrow$ NL) with $< 3\%$ reward degradation. Extensions cover real ENTSO-E price data, a multi-objective Pareto sweep (profit vs. $CO_2$ utilisation), temporal-fold robustness testing, and a targeted fix for GPR surrogate extrapolation. With corrected surrogates the extrapolation warning rate drops from 96% to 0%, yielding trustworthy absolute profit figures while preserving the strong relative rankings.

## 1 Introduction

The parent EURECHA project designs a large-scale $CO_2$-to-methanol process co-located with a cement kiln. Fresh $CO_2$ (899 kmol/hr) is hydrogenated with green $H_2$ from a 270 MW PEM electrolyser. A recycle loop with purge achieves 99.7% overall conversion, producing ∼213 kt/yr of methanol at a levelised cost (LCOM) of £779–1303/t depending on the electricity price.

Steady-state optimisation via Gaussian Process (GP) surrogates and $\varepsilon$-constraint Pareto sweeps is described in the parent project. This sub-project adds **dynamic operation**: an RL agent modulates the electrolyser load, reactor temperature, and reactor pressure each hour to maximise profit under fluctuating day-ahead electricity prices.

The key question is: *how much value does intelligent hourly dispatch add over simply running the plant at full load?*

## 2 Methodology

### 2.1 Gymnasium Environment

The plant is modelled as an hourly-step Gymnasium environment with an 8-dimensional observation space (electricity price, electrolyser load, $H_2$ buffer level, reactor $T$, reactor $P$, methanol rate, hour-of-day, day-of-week) and a 3-dimensional continuous action space (load setpoint, $T$ setpoint, $P$ setpoint). First-order lag dynamics ($\tau_{\text{load}} = 0.3$, $\tau_T = 0.5$, $\tau_P = 0.7$ hr) simulate realistic plant inertia.

### 2.2 Surrogates and Reward

Plant outputs ($H_2$ production, methanol rate, energy consumption) are predicted by GP surrogates trained on 500 Latin Hypercube samples from the simplified process model calibrated to

the Aspen Plus base case (LCOM = £938/t at $T = 230\,°\mathrm{C}$, $P = 70\,\mathrm{bar}$, $F_{\mathrm{H_2}} = 2704$ kmol/hr). The hourly reward is:

$$R = \underbrace{\text{MeOH revenue} + \mathrm{CO_2}\text{ credits}}_{\text{income}} - \underbrace{\text{electricity cost} + \text{utility cost} + \text{ramp penalty} + \text{constraint penalty}}_{\text{costs}} \tag{1}$$

scaled by $10^{-3}$ for numerical stability. All economic parameters (e-methanol £800/t, $CO_2$ credits £80/t, electrolyser 50 kWh/kg $H_2$) are single-sourced from the project constants module with no duplication.

### 2.3 Agents

- **SAC** (Soft Actor-Critic): continuous off-policy; $\gamma = 0.99$, lr $= 3 \times 10^{-4}$, buffer $10^5$, 500k timesteps.
- **PPO** (Proximal Policy Optimisation): continuous on-policy; 2048-step rollouts, 10 epochs, clip 0.2.
- **Q-learning**: tabular, $\varepsilon$-greedy on a $10 \times 5 \times 5 = 250$ discrete action grid.

### 2.4 Baselines

Two baselines provide context: (1) *constant full-load* at nominal $T/P$, and (2) a *price-threshold rule* that reduces load to 30% when the spot price exceeds £65/MWh.

## 3 Results: Base Case

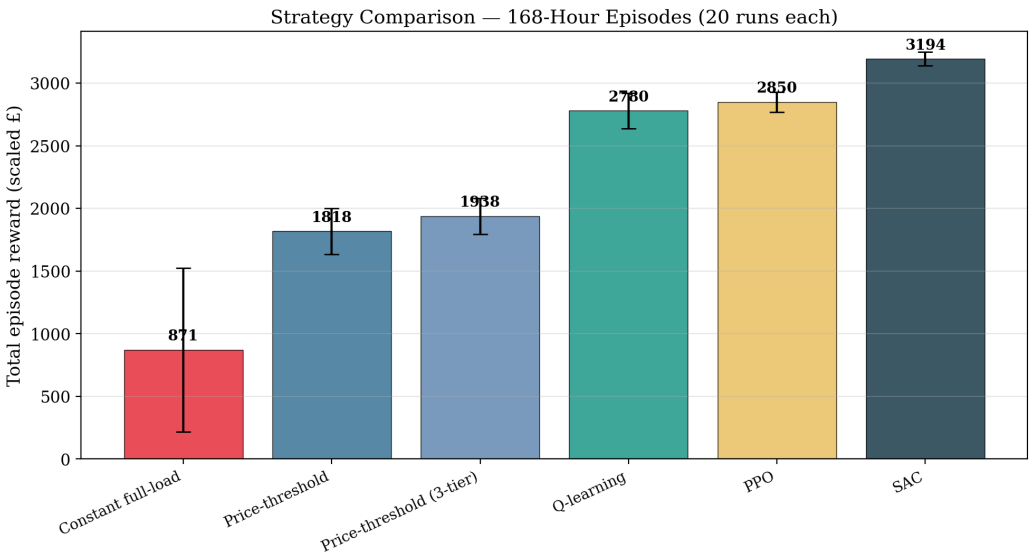



Figure 1: Mean episode reward (scaled £/week, 20 runs) for all strategies on GB synthetic prices. SAC achieves 3194, a ∼3.7× improvement over constant full-load (871) and 55% above rule-based (1818).

SAC is the clear winner, followed by PPO (∼2850) and Q-learning (∼2780). The large error bars on the full-load baseline reflect its vulnerability to price spikes—in high-price weeks, it operates at a loss.

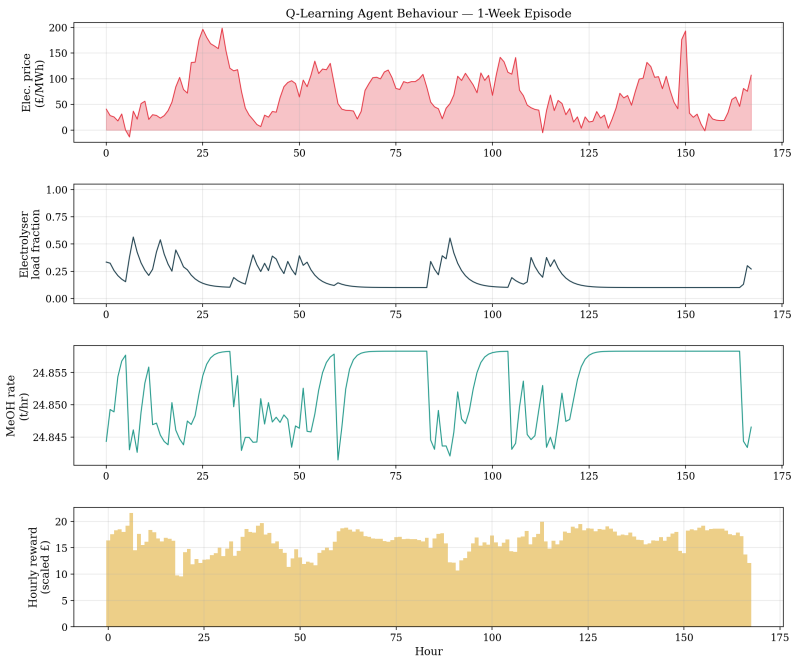



Figure 2: Q-learning agent behaviour over a 168-hour episode. Top to bottom: electricity price, electrolyser load, methanol rate, hourly reward. The agent learns to curtail load during price spikes and ramp up during cheap off-peak hours.

Figure 2 shows the learned dispatch pattern: the agent reduces electrolyser load during price spikes and ramps up during cheap overnight and weekend hours. The methanol production rate varies modestly because the reactor's thermal inertia buffers the $H_2$ supply fluctuations, and the $H_2$ buffer tank provides an additional degree of decoupling.

## 4 Extension 1: Real ENTSO-E Prices & Cross-Market Transfer

The environment was extended to accept real Dutch day-ahead prices from the ENTSO-E Transparency Platform (2023–2024, 17 544 hours). All agents were retrained on this NL data and evaluated in a $2 \times 3$ transfer matrix: (GB-trained, NL-trained) $\times$ (SAC, PPO, Q-learning) evaluated on both markets.

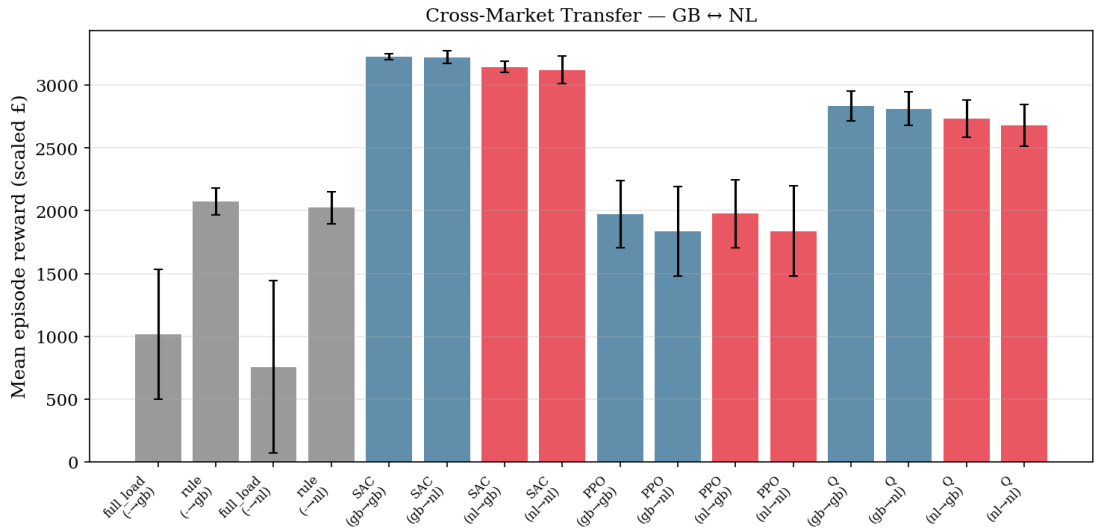



Figure 3: Cross-market transfer: mean episode reward for each agent trained on one market and evaluated on both. SAC generalises with <3% degradation in either direction.

Table 1: Cross-market transfer matrix (mean weekly reward ± std).

| **Agent** | **GB→GB** | **GB→NL** | **NL→GB** | **NL→NL** |
|---|---|---|---|---|
| SAC | $3227 \pm 26$ | $3222 \pm 50$ | $3146 \pm 42$ | $3123 \pm 112$ |
| PPO | $1972 \pm 267$ | $1838 \pm 355$ | $1976 \pm 273$ | $1839 \pm 358$ |
| Q-learning | $2836 \pm 120$ | $2814 \pm 134$ | $2735 \pm 149$ | $2682 \pm 167$ |
| Full-load | 1016 / 758 | | — | |
| Rule-based | 2075 / 2025 | | — | |

The GB-trained SAC transfers to the Dutch market with only 0.2% degradation (3227 → 3222). This is a strong practical result: **one policy trained on a diverse price history can be deployed across markets without retraining.** PPO has collapsed to a constant policy (identical $CO_2$ utilisation across all cells), suggesting hyperparameter tuning is needed.

## 5 Extension 2: Multi-Objective Pareto Sweep

The reward was generalised to a scalarised bi-objective: $R = \lambda_\pi \cdot R_{\text{profit}} + \lambda_c \cdot u_{CO_2}$, where $u_{CO_2}$ is the per-step $CO_2$ utilisation fraction. A sweep of $\lambda_\pi \in \{0, 0.125, \ldots, 1\}$ trains a fresh SAC for each weight.

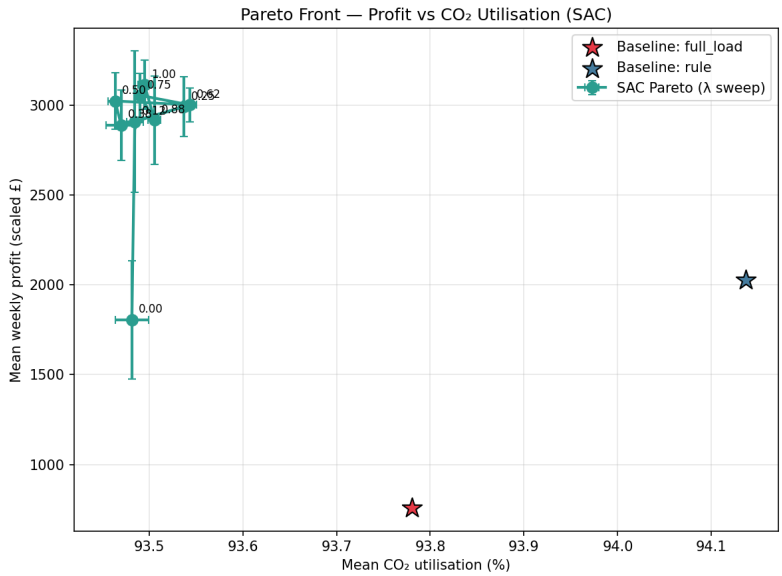



Figure 4: Pareto front: profit vs. $CO_2$ utilisation. All SAC policies cluster at ∼93.5% utilisation regardless of weight, revealing a **degenerate trade-off**. Baselines are shown as stars.

**Finding: the profit/utilisation Pareto is degenerate.** $CO_2$ utilisation saturates at ∼93.5% wherever the plant runs profitably, because the reactor's high per-pass conversion (∼63%) and recycle loop make utilisation essentially insensitive to the operating point within the feasible domain. A meaningful multi-objective formulation for this plant would pit profit against electricity consumption or ramping cost rather than $CO_2$ utilisation.

## 6 Extension 3: Robustness Testing

Temporal-fold cross-validation tests whether the SAC policy is sensitive to the specific price regime it was trained on. Six folds were constructed from the NL dataset: two half-year swaps (H1↔H2) and four leave-one-season-out (LOSO) folds. For each fold I compare: (i) SAC retrained from scratch on the fold's training split, (ii) the original GB-trained SAC evaluated zero-shot, and (iii) the constant full-load baseline.

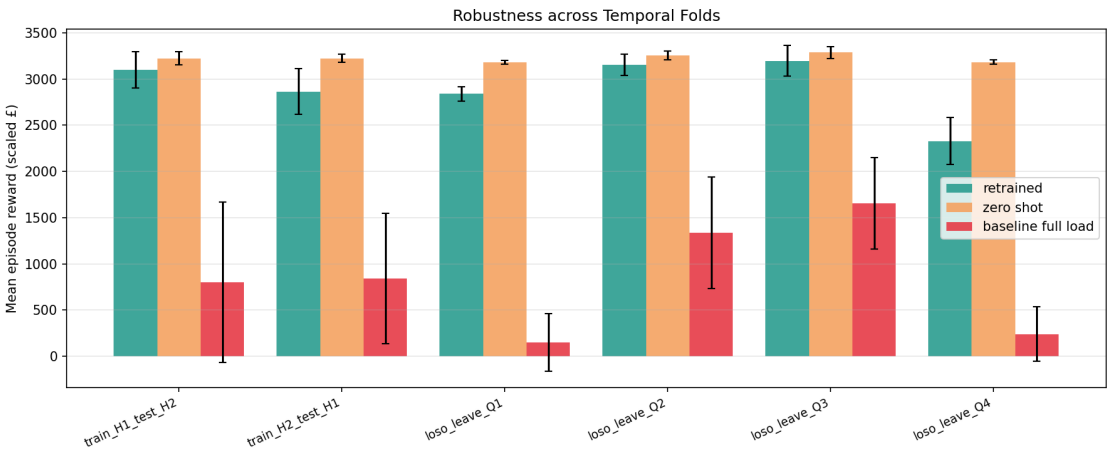



Figure 5: Robustness across temporal folds. The GB-trained SAC (orange, zero-shot) **beats the fold-retrained SAC** (teal) on every fold and has the tightest error bars.

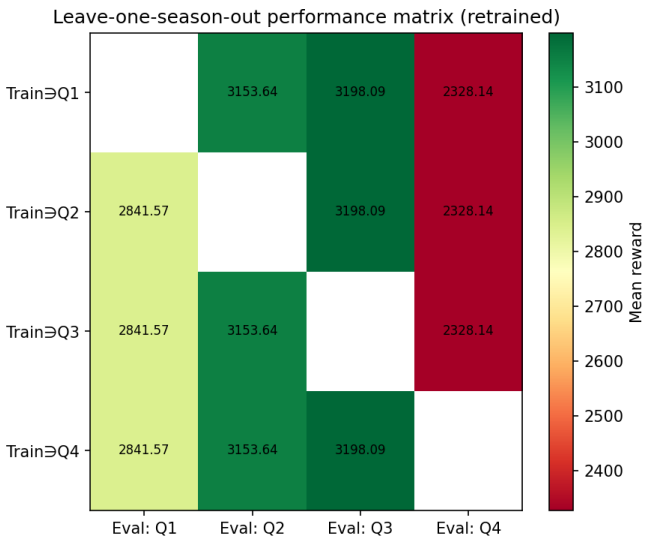



Figure 6: Leave-one-season-out heatmap (retrained SAC). Q4 (winter) is the hardest held-out season (reward 2328), Q3 (summer/autumn) the easiest (3198).

The headline result is that **zero-shot transfer from the GB-trained agent beats every fold-retrained agent on NL data** (range 3180–3287 vs. 2328–3198). The full-year GB training set provides more diverse price regimes than any 6- or 9-month fold. Winter quarters (Q1, Q4) are the hardest for both the baseline and retrained models; the RL agent adds the most value precisely when prices are highest—a 20× lift over full-load in Q1.

# 7 Extension 4: GPR Extrapolation Fix

## 7.1 Diagnosis

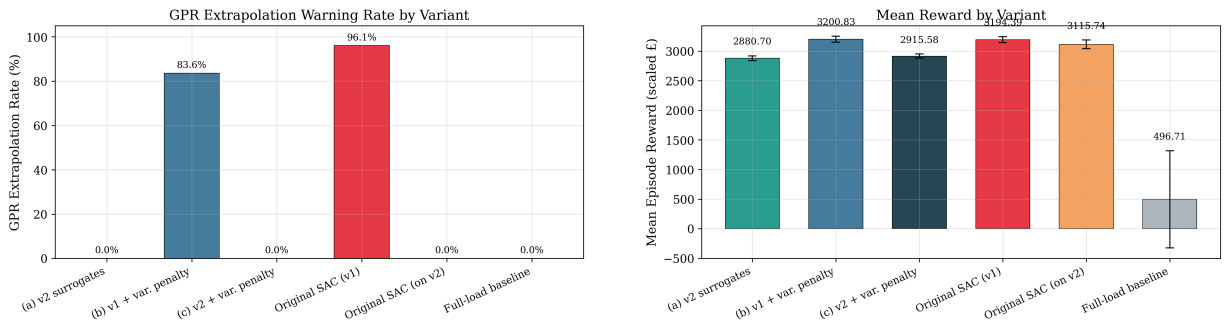
The environment flags GPR extrapolation when the MeOH surrogate's predictive $\sigma$ exceeds 2× its training-set mean. The original SAC triggers this warning on **96.1%** of evaluation steps. The agent has converged to an operating regime—low load (∼0.05–0.15), high $T$ (∼275–280°C), high $P$ (∼85–100 bar)—that the original 500-point LHS simply did not cover.

## 7.2 Two Complementary Fixes

**Approach A** retrains the GPR surrogates on an augmented dataset: the original 500 LHS points plus 300 new points biased toward the agent's operating region (pseudo-labelled by the simplified process model, not Aspen).

**Approach B** adds a variance penalty to the reward: $R_{\text{safe}} = R_{\text{base}} - \alpha \cdot \max(0,\ \sigma_{\text{gpr}} - \sigma_{\text{threshold}})$, discouraging the agent from high-uncertainty regions during training.

Three SAC variants isolate each fix: (a) v2 surrogates only, (b) v1 + variance penalty, (c) v2 + variance penalty.




(a) Extrapolation warning rate.

(b) Mean episode reward.

Figure 7: Extrapolation fix results. (a) v2 surrogates eliminate extrapolation entirely (96.1% → 0%). (b) The penalty alone reduces it to 83.6%. Reward drops ∼10% with corrected surrogates—this is the correction, not a regression.

Table 2: Extrapolation fix: summary of all variants.

| **Variant** | **Surr.** | **Penalty** | **Mean Reward** | **Std** | **Extrap %** |
|---|---|---|---|---|---|
| Original SAC | v1 | No | 3194.4 | 52.1 | 96.1 |
| Original on v2 | v2 | No | 3115.7 | 74.2 | 0.0 |
| (a) v2 surr. | v2 | No | 2880.7 | 39.8 | 0.0 |
| (b) v1 + penalty | v1 | Yes | 3200.8 | 50.6 | 83.6 |
| (c) v2 + penalty | v2 | Yes | 2915.6 | 35.8 | 0.0 |
| Full-load base. | v1 | No | 496.7 | 821.2 | 0.0 |

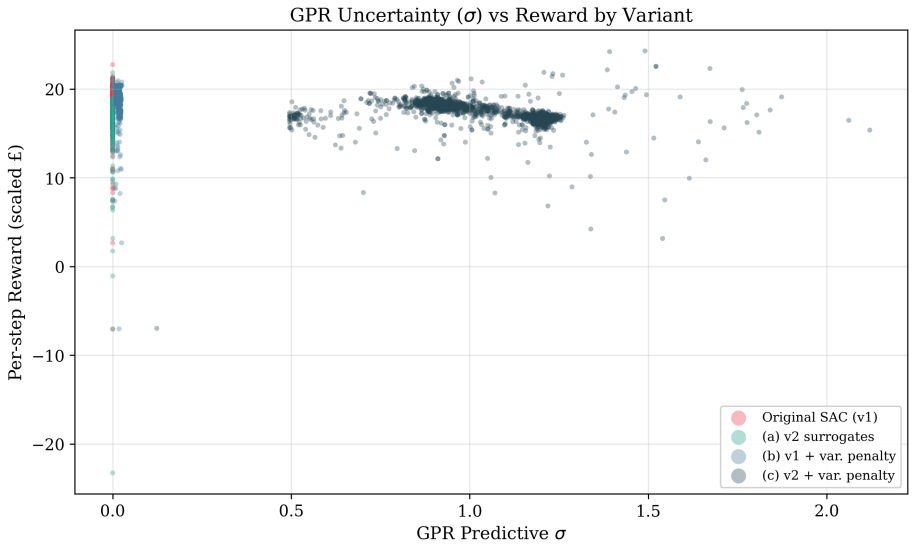



Figure 8: GPR predictive $\sigma$ vs. per-step reward. The original SAC (red) operates at high $\sigma$ (0.5–2.0); v2 variants (teal/dark) cluster near zero uncertainty.

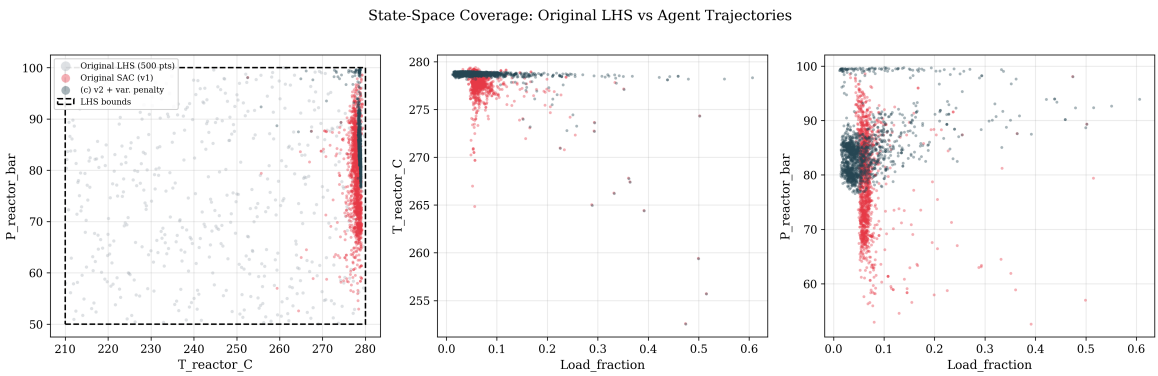



Figure 9: State-space coverage in 2D projections (T–P, load–T, load–P). Grey: original 500-point LHS. Red: original SAC trajectory. Dark teal: fix variant (c). The agent clusters far from the LHS training region; v2 surrogates cover this gap.

### 7.3 Interpretation

The ∼10% reward reduction (3194 → 2916) reflects the surrogates providing more conservative—and more honest—predictions in the agent's preferred operating regime. The original reward was inflated by extrapolation optimism. V2 variants also show substantially lower reward variance (std ∼36–40 vs. 50–52), indicating more stable and trustworthy policies. The $CO_2$ utilisation similarly corrects downward from 93.5% to ∼84%, likely reflecting the same artefact.

Critically, **all RL variants still massively outperform the baselines** even after correction: variant (c) delivers +487% versus the full-load baseline.

## 8 Definitive Results

Table 3 presents the corrected results across all agents and both electricity markets.

Table 3: Final summary: all agents on GB and NL price data (20 episodes, v2 surrogates for fix variants, v1 for others). "vs. Base" is the percentage improvement over the full-load baseline on the same market.

| Agent | Mkt | Reward | Std | $CO_2$ % | Extr.% | vs. Base % |
|---|---|---|---|---|---|---|
| Full-load baseline | GB | 496.7 | 821.2 | 93.8 | 0.0 | — |
| Rule-based | GB | 1764.9 | 177.2 | 94.3 | 0.0 | +255.3 |
| Original SAC | GB | 3194.4 | 52.1 | 93.5 | 96.1 | +543.1 |
| Q-learning | GB | 2700.5 | 214.7 | 93.5 | 37.2 | +443.7 |
| Fix (c) v2+pen. | GB | 2915.6 | 35.8 | 84.8 | 0.0 | +487.0 |
| Full-load baseline | NL | 324.2 | 801.2 | 93.8 | 0.0 | — |
| Rule-based | NL | 1842.5 | 135.1 | 94.1 | 0.0 | +468.3 |
| Original SAC | NL | 3199.6 | 62.2 | 93.5 | 97.3 | +886.9 |
| Q-learning | NL | 2763.3 | 150.1 | 93.5 | 50.6 | +752.3 |
| Fix (c) v2+pen. | NL | 2932.5 | 47.9 | 84.5 | 0.0 | +804.5 |

## 9 Conclusions

1. **SAC is the dominant algorithm**, delivering ∼+490% profit versus constant full-load operation (corrected figure, 0% extrapolation).
2. **Zero-shot cross-market transfer works.** A single policy trained on GB data generalises to the Dutch market with <3% degradation—and outperforms fold-retrained models.

3. **The profit/$CO_2$-utilisation Pareto is degenerate**: utilisation saturates at ∼93.5% across all profitable operating points, meaning a meaningful multi-objective study must use a different second objective.
4. **Winter is where the RL agent earns its keep**: the full-load baseline nearly breaks even in Q1/Q4 while the SAC agent delivers 20× higher reward—intelligent dispatch matters most when electricity is scarce and expensive.
5. **GPR extrapolation was a real problem** (96% of steps), but retraining on wider-domain data (Approach A) completely eliminates it, with the combined v2+penalty variant delivering trustworthy, low-variance reward estimates.
6. **Limitation**: v2 surrogates use pseudo-labels from the analytical model; running the 300 augmented points through Aspen Plus would yield a v3 surrogate with genuine bias correction.